

www.schmalzhaus.com/BigEasyDriver

An easy to use bipolar stepper motor driver
Use 4 wire, 6 wire or 8 wire stepper motors
From 0mA/phase to over 2A/phase
Defaults to 5V for Vcc (logic supply), settable to 3.3V
Supply 8V to 35V DC power input on JP1 or JP7
Do not connect or disconnect motor
while BigEasyDriver is powered

DEFAULT OPTIONS
Short JP10, or JP6 pins
to GND or Vcc to override

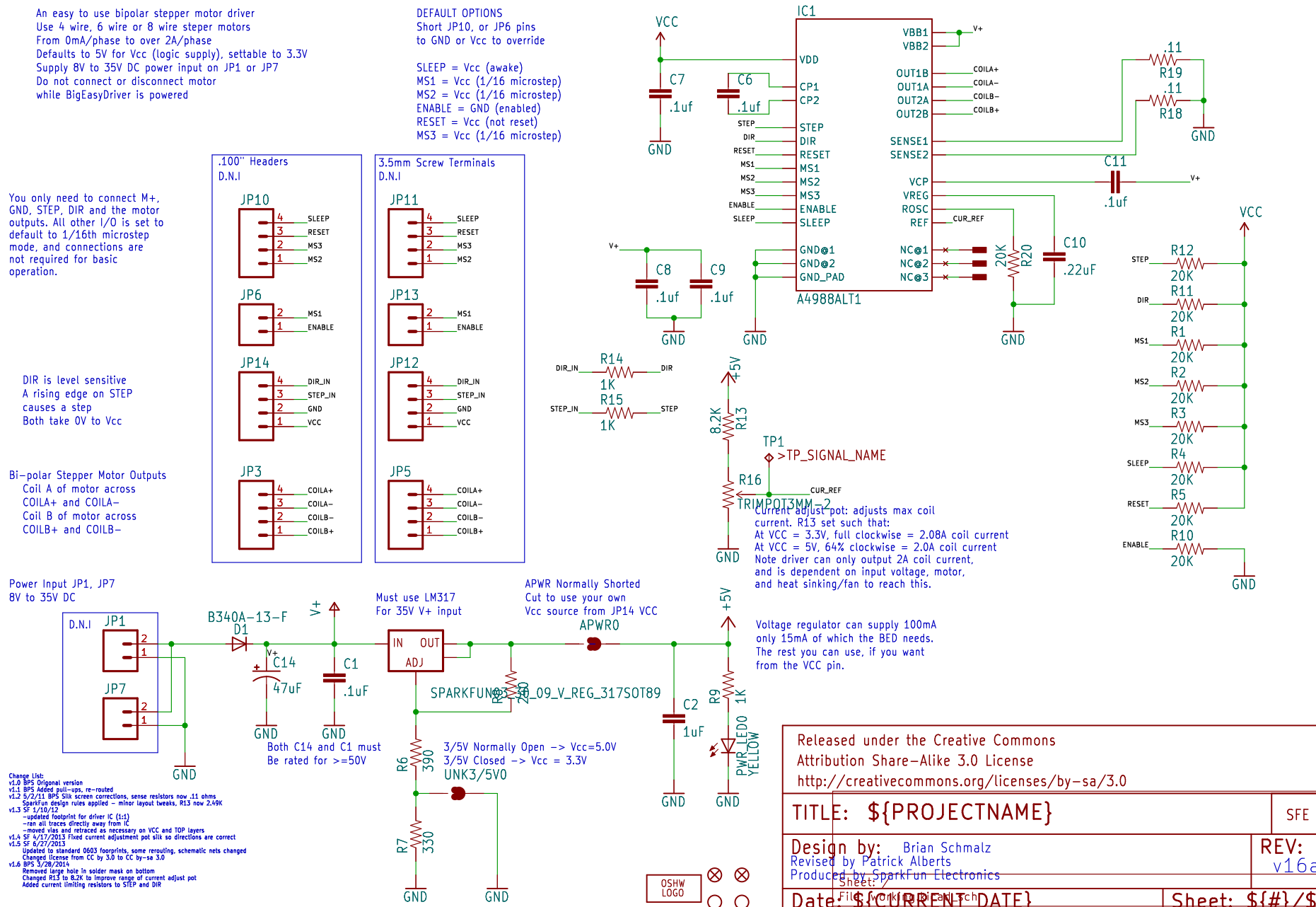
SLEEP = Vcc (awake)
MS1 = Vcc (1/16 microstep)
MS2 = Vcc (1/16 microstep)
ENABLE = GND (enabled)
RESET = Vcc (not reset)
MS3 = Vcc (1/16 microstep)

You only need to connect M+, GND, STEP, DIR and the motor outputs. All other I/O is set to default to 1/16th microstep mode, and connections are not required for basic operation.

DIR is level sensitive
A rising edge on STEP
causes a step
Both take 0V to Vcc

Bi-polar Stepper Motor Outputs
Coil A of motor across
COILA+ and COILA-
Coil B of motor across
COILB+ and COILB-

Power Input JP1, JP7
8V to 35V DC



NOTE: VCC is normally an OUTPUT. You do not need to supply power to the Big Easy Driver through VCC. The only power needed is through M+ (motor power).

Current adjust pot: adjusts max coil current. R13 set such that:
At VCC = 3.3V, full clockwise = 2.08A coil current
At VCC = 5V, 64% clockwise = 2.0A coil current
Note driver can only output 2A coil current, and is dependent on input voltage, motor, and heat sinking/fan to reach this.

Voltage regulator can supply 100mA only 15mA of which the BED needs. The rest you can use, if you want from the VCC pin.

- Change List:
- v1.0 BPS Original version
- v1.1 BPS Add pull-ups, re-routed
- v1.2 5/2/11 BPS Silk screen corrections, sense resistors now 11 ohms
- SparkFun design rules applied – mirror layout traces, R13 now 2.9K
- v1.3 SF 1/10/12
 - updated footprint for driver IC (1st)
 - ran all traces directly away from IC
 - moved vias and retraced as necessary on VCC and TOP layers
- v1.4 SF 6/17/2013 Added current adjustment pot on VCC so directions are correct
- v1.5 SF 6/27/2013
- Updated to standard 0603 footprints, some rerouting, schematic nets changed
- Change case from B to C by 3.0 to C by 3.0
- v1.6 BPS 5/28/2014
 - Removed large hole in solder mask on bottom
 - Changed R13 to 8.2K to improve range of current adjust pot
 - Added current limiting resistors to S1F and D1R

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TITLE: \${PROJECTNAME}

SFE

Design by: Brian Schmalz
Revised by Patrick Alberts
Produced by SparkFun Electronics

REV:
v16a

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Date: ${CURRENT_DATE}
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